

Power BI Notes :

1. Introduction to Power BI

- **Power BI** is a business analytics tool by Microsoft that helps visualize data and share insights.
- **Components:**
 1. **Power BI Desktop** – For creating reports and dashboards.
 2. **Power BI Service** – Cloud service for sharing and collaboration.
 3. **Power BI Mobile** – View reports on mobile devices.
 4. **Power BI Report Server** – On-premises report publishing.
- **Key Features:**
 - Connects to multiple data sources (Excel, SQL, Azure, APIs, etc.).
 - Data modeling and transformations (Power Query & DAX).
 - Interactive visualizations and dashboards.
 - Scheduled refresh and row-level security (RLS).
 - AI visuals and Q&A (natural language queries).

2. Power BI Workflow

1. **Connect to Data** – Import or connect live to data sources.
2. **Transform Data** – Clean, filter, merge using Power Query.
3. **Model Data** – Create relationships, hierarchies, and calculated columns/measures.
4. **Visualize Data** – Build charts, tables, maps, and dashboards.
5. **Share & Collaborate** – Publish reports to Power BI Service.

3. Data Sources

- **Excel & CSV**
- **SQL Server / Database** connections
- **Azure Data Lake / Azure SQL**
- **SharePoint Lists**
- **Web / APIs**

- **Other Services:** Salesforce, Dynamics 365, Google Analytics, etc.

4. Power Query (Data Transformation)

- **Key Functions:**
 - Remove columns/rows
 - Filter data
 - Merge queries (JOINS)
 - Append queries (UNION)
 - Pivot & Unpivot
 - Change data types
 - Conditional columns
 - Split columns
 - Fill up/down

5. Data Modeling in Power BI

1. What is Data Modeling?

Data modeling in Power BI is the process of **structuring data** to define how different tables relate to each other.

It helps Power BI understand how to combine data from multiple sources and allows you to perform calculations (like totals, averages, or comparisons) accurately using **DAX**.

2. Relationships

Relationships connect tables using common columns (keys).



Types of Relationships:

Type	Symbol	Description	Example
One-to-Many (1:*)	1:*	One value in one table relates to many in another.	One <i>Customer ID</i> → many <i>Orders</i>
Many-to-One (*:1)	*:1	Reverse of One-to-Many; used when connecting a fact table to a dimension table.	Many <i>Sales Rows</i> → one <i>Product</i>
Many-to-Many (:)	:	When both tables have non-unique values. Needs a bridge/intermediate table.	<i>Students</i> ↔ <i>Courses</i> (many students can join many courses)

Cardinality

- Cardinality defines the **uniqueness** of data between tables.
- It impacts **DAX functions** like CALCULATE(), RELATED(), and context propagation.

Cross Filter Direction

- **Single:** Filters flow in one direction (usually from dimension → fact table).
- **Both:** Allows filters to flow both ways; used in complex models.

3. Keys in Data Modeling

Key Type	Description	Example
Primary Key	A unique identifier in a table (no duplicates).	CustomerID in Customer table
Foreign Key	A reference to a primary key in another table.	CustomerID in Orders table links to Customer table

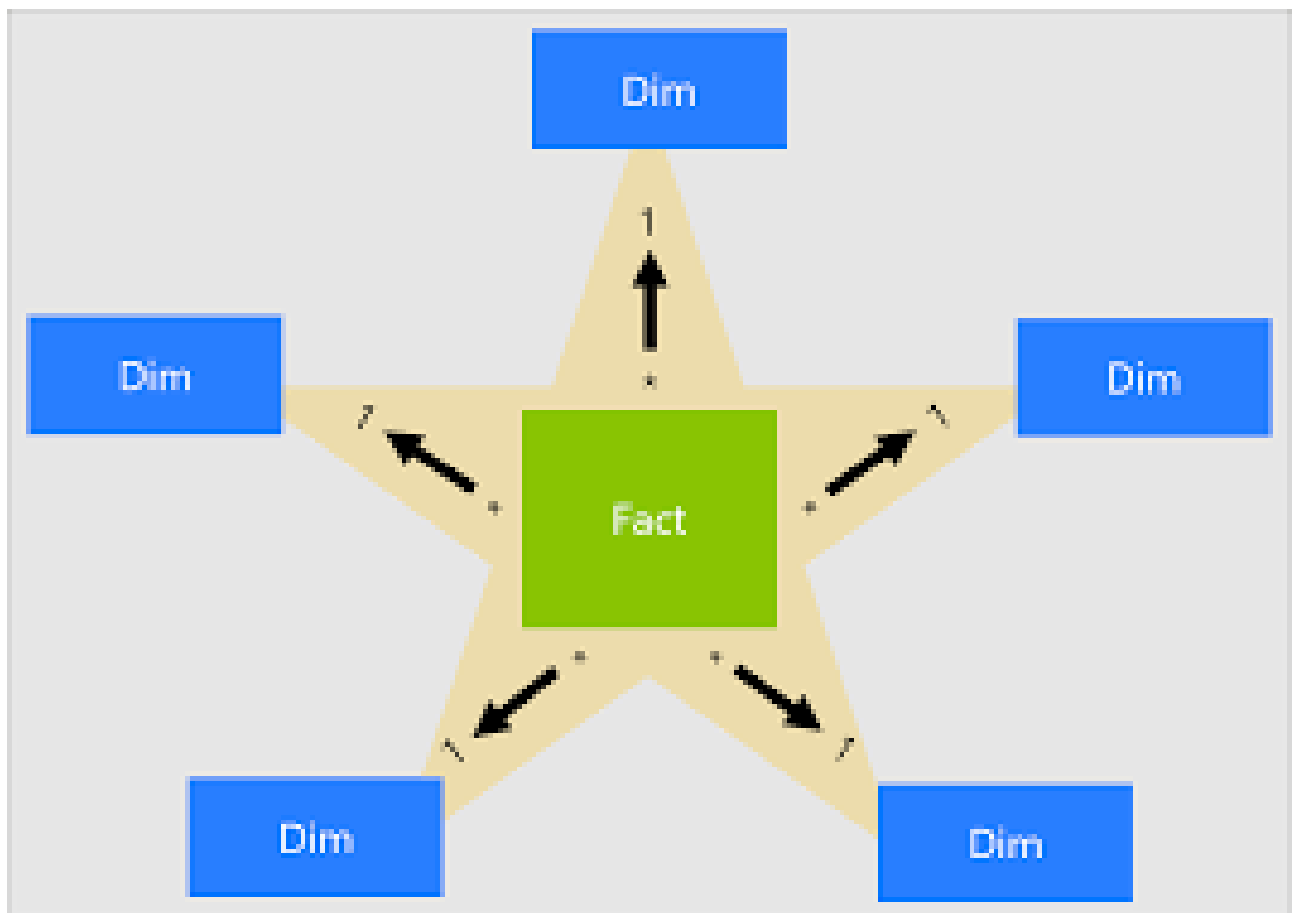
The relationship is built on matching Primary Key ↔ Foreign Key.

4. Star Schema vs Snowflake Schema

Star Schema

- Structure:

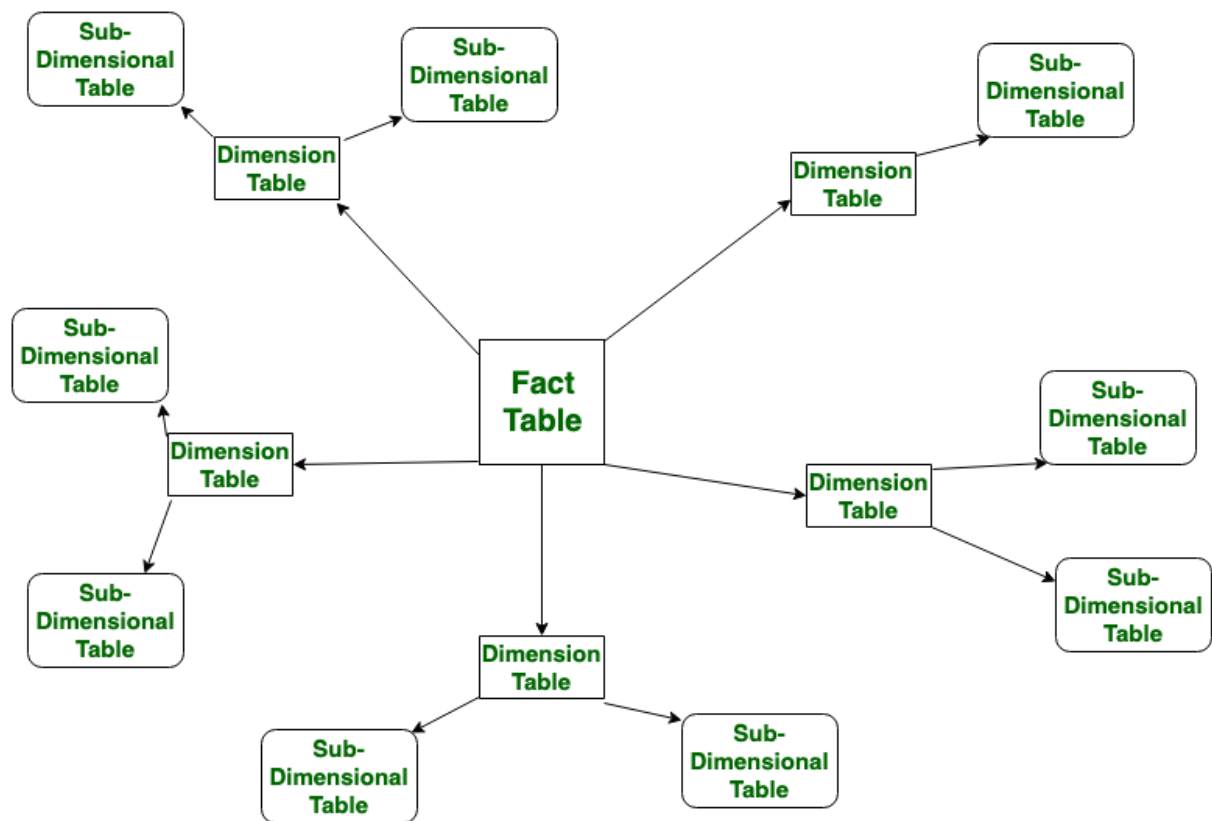
- **1 Fact Table** (contains numeric, measurable data — like sales, revenue, quantity)
- **Multiple Dimension Tables** (contain descriptive attributes — like customer name, product category, date)
- **Simple and Fast**
- Easy to visualize relationships
- Best for Power BI performance



Snowflake Schema

- Extension of star schema.
- Dimension tables are **normalized** (split into sub-tables).
- **More complex** → slower performance.

Use Case: When dimension tables are large and repeated data must be avoided.



5. Fact and Dimension Tables

Table Type	Contains	Example Columns	Example
Fact Table	Quantitative data (measures)	SalesAmount, Quantity	Sales
Dimension Table	Descriptive attributes	CustomerName, ProductName	Customer, Product

6. Example Relationship Setup

Tables:

- Sales (Fact): OrderID, CustomerID, ProductID, Amount
- Customer (Dimension): CustomerID, Name, City
- Product (Dimension): ProductID, ProductName, Category

Relationships:

- Customer[CustomerID] → Sales[CustomerID] (1:*)
- Product[ProductID] → Sales[ProductID] (1:*)

Now you can use DAX functions like:

. Best Practices for Data Modeling

Use **Star Schema** whenever possible.

Create **unique keys** for each table.

Avoid **many-to-many** relationships unless required.

Define **clear relationship directions** (usually single).

Remove **unused columns** and ensure proper **data types**.

Keep **fact tables wide** (more measures) and **dimension tables long** (more rows).

6. DAX (Data Analysis Expressions)

Purpose:

Used to create **calculated columns**, **measures**, and **calculated tables** in Power BI for advanced analysis and dynamic calculations.

Basic Functions:

Function	Description	Example
SUM()	Returns the total of a numeric column.	SUM(Sales[Amount])
AVERAGE()	Returns the mean of a numeric column.	AVERAGE(Sales[Profit])
COUNT()	Counts numeric values only.	COUNT(Sales[Quantity])
COUNTA()	Counts all non-blank values.	COUNTA(Customer[Name])
COUNTROWS()	Counts the number of rows in a table.	COUNTROWS(Sales)
MAX() / MIN()	Returns the highest/lowest value.	MAX(Sales[Amount])

Logical Functions:

Function	Description	Example
IF()	Checks a condition and returns a value if true or false.	IF(Sales[Profit] > 0, "Profit", "Loss")
SWITCH()	Replaces multiple IFs, checks against multiple cases.	SWITCH(TRUE(), [Score] >= 90, "A", [Score] >= 80, "B", "C")

Filter Functions:

Function	Description	Example
CALCULATE()	Changes the filter context to perform dynamic calculations.	CALCULATE(SUM(Sales[Amount]), Region[Name] = "East")
FILTER()	Returns a filtered table that can be used inside CALCULATE.	FILTER(Sales, Sales[Amount] > 1000)

Time Intelligence Functions:

Function	Description	Example
TOTALYTD()	Calculates year-to-date values.	TOTALYTD(SUM(Sales[Amount]), Date[Date])
SAMEPERIODLASTYEAR()	Compares data from the same period last year.	CALCULATE(SUM(Sales[Amount]), SAMEPERIODLASTYEAR(Date[Date]))
DATEADD()	Shifts dates by a specific interval (days, months, years).	CALCULATE(SUM(Sales[Amount]), DATEADD(Date[Date], -1, YEAR))

Measure vs Calculated Column

Feature	Measure	Calculated Column
Calculation Type	Calculated on the fly (runtime)	Calculated row-by-row
Storage	Not stored in table	Stored in table
Performance	Faster (less memory)	Slower (more memory)
Context	Filter context	Row context
Usage	Used in visuals, cards, KPIs	Used in data model relationships or filters

7. CHART AND VISUALIZATION

1. Clustered Column Chart

Definition: A chart that displays data with vertical bars grouped by category to compare values side by side.

X-Axis: Categories (e.g., Product, Region)

Y-Axis: Numeric values (e.g., Sales, Profit)

Legend: Optional (e.g., Year)

Use: Compare values across categories side by side.

Example: Sales by Region for different years.

2. Stacked Column Chart

Definition: A vertical bar chart where values are stacked on top of each other to show parts of a whole.

X-Axis: Categories

Y-Axis: Total value (stacked by sub-category)

Legend: Sub-category (e.g., Product Type)

Use: Show contribution of each category to the total.

Example: Total sales by region, divided by product type.

3. 100% Stacked Column Chart

Definition: Displays percentage contributions of sub-categories that stack up to 100%.

X-Axis: Categories

Y-Axis: Percentage (0–100%)

Legend: Sub-category

Use: Compare percentage contribution of each sub-category.

Example: Market share percentage by brand in each region.

4. Clustered Bar Chart

Definition: A horizontal version of a clustered column chart used to compare long category names.

X-Axis: Numeric value (e.g., Sales)

Y-Axis: Categories (horizontal layout)

Legend: Optional

Use: Compare long category names horizontally.

Example: Sales by Country.

5. Stacked Bar Chart

Definition: Displays horizontal bars divided into segments representing sub-categories stacked together.

X-Axis: Total numeric value

Y-Axis: Category

Legend: Sub-category

Use: Same as stacked column but horizontal.

Example: Total sales by department (stacked by product type).

6. 100% Stacked Bar Chart

Definition: A horizontal chart showing relative percentage contributions that total 100%.

X-Axis: Percentage

Y-Axis: Category

Legend: Sub-category

Use: Compare relative percentage distribution.

Example: Customer type (New vs Returning) per region.

7. Line Chart

Definition: Displays trends over time by connecting data points with a line.

X-Axis: Continuous variable (usually Date/Time)

Y-Axis: Numeric value (e.g., Sales)

Legend: Series (e.g., Product Category)

Use: Show trend over time.

Example: Monthly revenue trend.

8. Area Chart

Definition: A line chart with the area below the line filled to emphasize volume or magnitude.

X-Axis: Time or Category

Y-Axis: Value (filled area under line)

Legend: Series

Use: Emphasize volume or total change over time.

Example: Cumulative revenue trend.

9. Stacked Area Chart

Definition: Displays cumulative totals over time by stacking multiple area series.

X-Axis: Time

Y-Axis: Stacked total values

Legend: Series (categories)

Use: Show how each category contributes over time.

Example: Sales by region over months.

10. Line and Clustered Column Chart

Definition: Combines a column and line chart to compare two related measures on shared axes.

Shared X-Axis: Category (e.g., Month)

Y-Axis (Left): Column values

Y-Axis (Right): Line values

Use: Compare two related metrics.

Example: Revenue (columns) vs Profit Margin (line).

11. Waterfall Chart

Definition: Shows how an initial value is affected by positive and negative changes leading to a final total.

X-Axis: Categories/Events

Y-Axis: Cumulative value

Use: Show how starting value changes with positive and negative values.

Example: Profit bridge from revenue to net income.

12. Scatter Chart

Definition: Plots points for two numeric variables to show relationships, patterns, or correlations.

X-Axis: Numeric variable 1

Y-Axis: Numeric variable 2

Size: Optional (e.g., sales volume)

Color/Legend: Category

Use: Show relationships and correlations.

Example: Advertising Spend vs Revenue.

13. Pie Chart

Definition: Displays proportions of a whole using slices of a circle.

Legend: Category

Values: Numeric value

Use: Show proportion of categories to a whole.

Example: Market share by company.

14. Donut Chart

Definition: A circular chart similar to a pie chart but with a hollow center to show totals or percentages.

Legend: Category

Values: Value

Use: Same as pie chart; center space can show total.

Example: % contribution of each product to total sales.

15. Map

Definition: Plots data points geographically using latitude, longitude, or region fields.

Location Field: City, Country, Lat/Long

Size: Value (bubble size)

Color: Category or value intensity

Use: Show geographic distribution.

Example: Sales by location.

16. Filled Map

Definition: Colors entire geographic areas based on data values (heat or intensity).

Location Field: Geographic area

Color Saturation: Value

Use: Show density or intensity across regions.

Example: Population density by state.

17. Shape Map

Definition: Uses custom geographic or region shapes (GeoJSON) to display value intensity.

Location Field: Custom shape/region (GeoJSON)

Color: Value intensity

Use: Custom region-based visualization.

Example: Sales by custom sales territory.

18. Card

Definition: Displays a single KPI or metric in large text format.

Value: Single number (KPI)

Use: Show a single key metric.

Example: Total Revenue = ₹1.2M

19. Multi-row Card

Definition: Displays multiple KPIs or fields in a compact, readable layout.

Fields: Multiple values

Use: Display several KPIs together.

Example: Total Sales, Profit, and Customers.

20. Table

Definition: Displays raw, detailed data in rows and columns similar to an Excel sheet.

Columns: Fields (text/numeric)

Use: Show detailed data, raw format.

Example: List of transactions with amount, date, customer.

21. Matrix

Definition: A pivot-table style visual showing summarized data across rows and columns.

Rows: Categories

Columns: Categories (like pivot)

Values: Aggregated measures

Use: Cross-tab style summary.

Example: Region (rows) × Product (columns) showing sales.

22. Decomposition Tree

Definition: A visual that breaks down a measure hierarchically to explore root causes.

Analyze Field: Main measure (e.g., Sales)

Explain By: Dimensions (Region → Product → Month)

Use: Root cause or hierarchical breakdown.

Example: Why did profit drop this month?

23. Key Influencers

Definition: An AI visual that identifies key factors influencing a selected metric.

Analyze: Dependent variable (target)

Explain By: Independent variables (factors)

Use: Find what drives changes in a metric.

Example: What influences customer churn?

24. Smart Narrative

Definition: Automatically generates text-based summaries and insights from visuals.

Inputs: Any visual/measure

Output: Auto-generated text insights

Use: Auto summaries (“Sales grew 12% vs last year”).

25. Gauge

Definition: A dial-like visual showing progress toward a specific target.

Value: KPI measure

Target: Goal value

Min/Max: Range

Use: Show performance against goal.

Example: Sales = ₹90K (Target ₹100K).

26. KPI

Definition: Shows a key metric’s performance, target, and trend direction.

Indicator: Current value

Trend Axis: Date/Time

Target: Goal value

Use: Track performance and direction (up/down).

Example: Profit margin trend vs target.

27. Slicer

Definition: An interactive filter used to control data displayed across visuals.

Field: Any category/date

Use: Add interactive filter.

Example: Filter report by Year or Product.

28. R / Python Visual

Definition: Runs custom R or Python scripts to generate advanced visuals or models.

Inputs: Data fields

Script Output: Custom chart (R or Python code)

Use: Advanced visuals and ML outputs.

Example: Regression line, clustering plot.

29. Q&A Visual

Definition: Allows users to ask natural language questions and generates dynamic visuals.

Input: Natural language question (typed)

Output: Dynamic visual answer

Use: Interactive data exploration.

Example: "Show sales by region in 2024."

30. Power Automate Visual

Definition: Integrates with Power Automate to trigger workflows from Power BI reports.

Trigger: Button inside report

Use: Run workflow (send email, update record, etc.)

Example: Send alert when profit < target.

1. Goal (Power BI Goals in Power BI Service)

Definition:

Power BI **Goals** is a performance tracking feature inside Power BI Service (in "Scorecards"). It helps organizations **define, track, and manage key business objectives (KPIs)** in real time.

Key Features:

- Create scorecards with measurable goals and sub-goals.
- Link each goal directly to Power BI report data for live tracking.

- Supports status indicators like *On Track*, *At Risk*, or *Behind*.
- Enables collaboration by assigning owners and due dates.

Use:

Track business performance such as revenue targets, sales goals, or project milestones.

Example:

Track “Monthly Sales Target = ₹5,00,000” with automatic progress updates from sales reports.

2. Paginated Report

Definition:

Paginated Reports are **pixel-perfect, print-ready reports** in Power BI, designed for detailed layouts like invoices, statements, or forms.

They use **Power BI Report Builder (RDL format)** and can display **large tables across multiple pages** — hence the name *paginated*.

Key Features:

- Perfect for exporting or printing (PDF, Excel).
- Supports headers, footers, and precise layout control.
- Works well for operational reporting and financial summaries.

Use:

Generate detailed, formatted reports that can be printed or shared.

Example:

Monthly employee salary statements or customer billing reports.

3. ArcGIS for Power BI

Definition:

ArcGIS is an **advanced mapping and spatial analysis visual** integrated within Power BI (by Esri). It enables **geographic analysis** with layered map visualizations beyond the basic map visual.

Key Features:

- Add multiple layers like heat maps, reference layers, and demographic data.
- Perform spatial analysis such as clustering or drive-time areas.
- Uses **Esri’s geographic intelligence** for location-based insights.

Use:

Analyze data geographically to understand spatial trends or density.

Example:

Visualize store sales by region and identify high-performing areas within 10 km of a city center.

4. Power Apps Visual for Power BI**Definition:**

This visual embeds a **Power Apps app** directly inside a Power BI report. It allows users to **view, edit, or add data** interactively — bridging **data visualization and data entry**.

Key Features:

- Integrates real-time data input into Power BI.
- Allows users to take actions (e.g., update or submit data).
- Two-way connection between Power BI and Power Apps datasets.

Use:

Enable users to **edit or submit data** directly within reports.

Example:

A sales dashboard where users can update deal status or add new customer details without leaving Power BI.

5. Power Automate for Power BI**Definition:**

The **Power Automate visual** integrates Power BI with **Power Automate flows (Workflows)**. It lets users trigger **automated actions** (like notifications, approvals, or updates) directly from a report.

Key Features:

- Create automated workflows triggered by button clicks or data changes.
- Send alerts, update databases, or integrate with Teams, Outlook, etc.
- Simplifies repetitive business processes.

Use:

Automate workflows from Power BI insights.

Example:

When profit < target, send an automatic email alert to the finance manager.

6. Decomposition Tree (AI Visual)

Definition:

An **AI-powered visual** in Power BI that **breaks down a measure** (like Sales or Profit) **into hierarchical components** to explore **root causes** or detailed analysis.

Key Features:

- Interactive drill-down analysis.
- AI can automatically suggest the next best dimension for analysis.
- Useful for *why-analysis* and performance breakdowns.

Use:

Identify the reasons behind an increase or decrease in key metrics.

Example:

Break down “Profit” → by “Region” → “Product Category” → “Month” to find where losses occurred.

7. Key Influencers (AI Visual)

Definition:

An AI visual that identifies **factors influencing a target metric** by analyzing relationships between variables.

Key Features:

- Uses machine learning to detect which fields affect your outcome most.
- Displays key drivers and their impact direction (positive/negative).
- Helps in predictive or diagnostic analysis.

Use:

Find what impacts a specific result, like customer satisfaction or churn.

Example:

“What influences customer churn?” → Subscription duration and support quality might be top influencers.

8. Smart Narrative (Narrative Visual)

Definition:

Automatically generates **text-based insights and summaries** from your visuals and measures. It uses natural language generation to explain data trends and key points.

Key Features:

- Converts visuals into readable summaries.
- Dynamically updates when underlying data changes.
- Provides quick data storytelling without manual writing.

Use:

Summarize dashboards automatically with textual explanations.

Example:

“Total Sales increased by 12% compared to last month, mainly driven by North Region performance.”

8. Filters & Slicers

Purpose:

Filters and slicers are used to refine and focus data in reports. They help control **what data is displayed** in visuals.

1. Visual-Level Filter

- **Applies to:** A single visual only.
- **Use Case:** When you want to show data for a specific product, region, or category in one chart without affecting others.
- **Example:**
A bar chart shows sales only for “Electronics,” while the rest of the page shows all categories.

2. Page-Level Filter

- **Applies to:** All visuals on the same report page.
- **Use Case:** When you want to view a specific subset of data (e.g., year 2024) across all visuals on that page.
- **Example:**
Filtering the page to show only “North Zone” sales—every chart updates accordingly.

3. Report-Level Filter

- **Applies to:** The entire report (all pages).
- **Use Case:** When you want every visual and page to focus on the same filtered dataset.
- **Example:**
Filtering the whole report to show only “Active Customers.”

4. Slicers

- **Definition:**

Visual elements that allow **end users to interactively filter** data by clicking or selecting values.

- **Purpose:**

Give users **control** to dynamically change what data is shown in visuals.

Types of Slicers:

1. **List Slicer:**

- Displays all items vertically.
- Example: Select one or multiple product names.
- **Best for:** Few items.

2. **Dropdown Slicer:**

- Collapsible version of list slicer.
- **Best for:** Many items; saves space.

3. **Date Slicer:**

- Allows filtering by date range.
- Example: Show sales from 1-Jan-2024 to 30-Jun-2024.

4. **Relative Date Slicer:**

- Filters data based on dynamic time ranges.
- Example: “Last 7 days,” “Last 1 month,” “Next quarter.”

5. **Hierarchy Slicer:**

- Enables filtering across multiple levels (e.g., Year → Quarter → Month → Day).
- **Example:** Drill down from “2023” → “Q4” → “December.”

9. Row-Level Security (RLS)

Purpose:

- Restrict **data access** so that users see only the data relevant to them.
- Ensures **confidentiality** and **role-based access** within the same report.

Steps to Implement RLS

1. Create Roles in Power BI Desktop

- Go to **Modeling** → **Manage Roles**.
- Create roles like Sales_Manager, Regional_Head, or Employee.
- Each role represents a **group of users** with similar data access needs.

2. Define DAX Filter for Role

- Apply a **DAX expression** to filter rows for each role.
- Example:
- [Region] = "North"

This ensures that users in the North role see only North region data.

- You can also use dynamic filtering with USERNAME() or USERPRINCIPALNAME() to filter data per logged-in user:
- [UserEmail] = USERPRINCIPALNAME()

3. Publish Report → Assign Users in Power BI Service

- Publish the PBIX file to **Power BI Service**.
- Go to **Dataset** → **Security**.
- Assign **actual users or groups** to each role.
- When users open the report, they see only the rows defined by their assigned role.

Tips:

- Test roles in **Power BI Desktop**:
Modeling → View as Role to simulate what users will see.
- Combine RLS with **slicers and filters** for enhanced control.
- Dynamic RLS using DAX is better for large organizations with many users.

10. Power BI Service

Purpose:

- Power BI Service is the **cloud-based platform** for sharing, collaborating, and managing Power BI reports and dashboards.

Key Components

1. Workspaces

- Collaborative environment for **teams or projects**.
- Users can **create, edit, and manage** reports, datasets, and dashboards together.
- Example: Marketing team workspace for all campaign reports.

2. Apps

- Packaged collection of **reports and dashboards** that can be shared with **end users**.
- Easier to distribute insights without giving edit access.
- Example: Sales Performance App for managers.

3. Dashboards

- Single-page, **interactive view** combining visuals from **multiple reports**.
- Allows quick insights and monitoring of KPIs.
- Example: CEO dashboard showing sales, revenue, and customer metrics from different reports.

4. Data Gateway

- Bridges **on-premises data sources** with Power BI Service.
- Enables **refreshing data** from local databases without moving the data to the cloud.
- Types:
 - **Personal Gateway:** For individual use.
 - **Enterprise Gateway:** For multiple users/teams.

5. Scheduled Refresh

- Automatically updates datasets and dashboards at **set intervals**.
- Ensures **real-time or near real-time data** is visible in reports.
- Example: Refresh daily at 6 AM to show yesterday's sales data.